Merge Sort and Divide-and-Conquer: Solutions

02-E05 Complete stable MERGE

The comparisons at (i,j) are (1,1), selecting 0a; (2,1), selecting 1d; (2,2), selecting 3b on equality; (3,2), selecting 3e; and (3,3), selecting 6f. Output prefixes are [0a], [0a,1d], [0a,1d,3b], [0a,1d,3b,3e], and [0a,1d,3b,3e,6f]. R is exhausted; append 7c. There are 5 comparisons, 6 buffer copies, and 6 output writes.

Right priority changes output to [0a,1d,3e,3b,6f,7c]. It remains ordered and a permutation, satisfying sorting without stability, but fails stable sorting. Five comparisons attain the m−1 bound.

02-E06 Sentinels and aliases have assumptions

The buffers become [M_real,M_sentinel] and [M_real,M_real,M_sentinel]. With left priority, output 1 selects L[1], output 2 selects the left sentinel, and output 3 attempts L[3], outside L. A sentinel equal to a legal key may be selected prematurely.

For views of [2,4,1,3], choosing right-head 1 first writes A[1]=1, changing the unread left-head view from 2 to 1. The next selection reads this overwritten 1 and duplicates data. Use independent copies and length tests; after one side empties, read only the other. A sentinel alternative needs a tag ranking above all real records, not merely the greatest legal key.

02-E07 Odd lengths and peak space

The root is (1,3,5). Its left call is (1,2,3), whose left call (1,1,2) merges to [2b,6a]. The three-record merge gives [2b,5c,6a]. The right call (4,4,5) gives [1d,4e]. The root gives [1d,2b,4e,5c,6a]. Merge sizes 2,3,2,5 total 12 buffer copies and 12 writes.

The largest individual merge buffers 5 records. Child buffers are released before parent merging, giving peak buffer demand 5, distinct from cumulative 12. The deepest live call chain has lengths 5→3→2→1, four calls. Generally stack space is Θ(log n) and record-buffer space Θ(n).

02-E08 Exact recurrence solutions

Let h=lg n. Every internal depth of U costs 5n, and leaves cost 3n, giving $U(n) = 5n \lg n + 3n$, hence U(8)=120+24=144. At depth ℓ, W costs $2^\ell(n/2^\ell - 1) = n - 2^\ell$. Summing depths 0 through h−1 gives $W(n) = n \lg n - n + 1$, hence W(8)=17. Substitution and the n=1 base cases also establish induction.

Merge Sort and Divide-and-Conquer: Solutions (continued)

Actual MERGE comparisons depend on exhaustion positions. W uses m−1 for each merge, a worst-case recurrence. Sorted eight-record input uses 4+4+4=12 comparisons rather than 17. Buffer copies and writes still total 8 per internal level; fewer comparisons do not remove all other work.

02-E09 Θ bounds beyond powers of two

Balanced floor/ceiling splits give depth-ℓ interval sizes between $\lfloor n/2^\ell \rfloor$ and $\lceil n/2^\ell \rceil$, until singleton termination. Height is at most ceil(lg n). At each depth the unfinished intervals are disjoint and have total size at most n. Internal cost at most a fixed c_U times size gives O(n log n).

For $\ell < \lfloor \lg n \rfloor$, every interval has size at least 2, so all are internal and cover n records. Each such level costs at least $c_L n$. There are floor(lg n) such levels, at least lg n/2 for n≥2, giving Ω(n log n). Exactly n singleton leaves cost Θ(n) in total and do not change the main order. Sizes 0 and 1 cost Θ(1) separately. The argument uses actual intervals, without an unproved monotonicity assumption.

02-E10 Model changes and an early-exit variant

Sorted input causes no insertion shifts, giving Θ(n). Standard MERGE still copies and writes every internal interval, giving Θ(n log n). For the variant, after sorting both halves, A[q] is the left maximum and A[q+1] the right minimum. If their comparison succeeds, concatenation is ordered; equal left records already precede equal right records, preserving stability. On sorted input every merge is skipped. The recursion tree has n leaves and n−1 internal nodes with constant checking work, giving Θ(n). Worst-case time remains O(n log n), since skipping only removes work; interleaved inputs can retain merging costs.

If comparisons require Θ(L) character reads in the worst case, charge each actual comparison accordingly instead of treating every guard as constant. Reference moves may still cost O(1); indiscriminately multiplying every operation by L is not an exact cost model.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study